



INTRODUCTION

- Science refers to a systematic and organized body of knowledge in any area of inquiry that is acquired using “the scientific method”
- It can be grouped into two broad categories:
 - **natural science** (i.e., the science of naturally occurring objects or phenomena, such as light, matter, earth, celestial bodies, or the human body)
 - **social science** (i.e., the science of people or collections of people, such as groups, firms, societies, or economies, and their individual or collective behavior).

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- In today's class, we will focus on the social sciences ...

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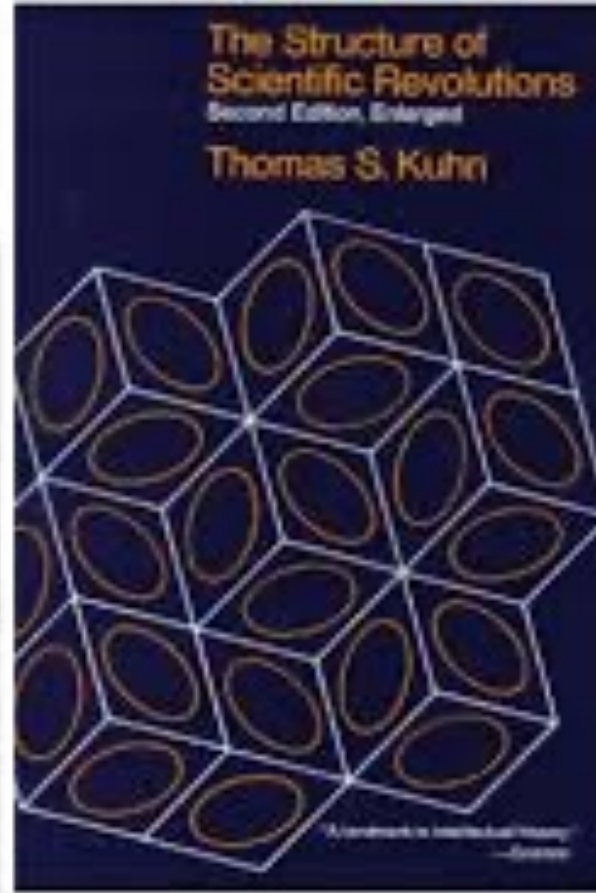


- 1. Explain the idea of paradigms in science. How might different paradigms offer different explanations for the same observed phenomena?

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- Our design and conduct of research is shaped by our *mental models* or *frames of references* that we use to organize our reasoning and observations.
- This *mental models* or *frames* (*belief systems*) are called **paradigms**.

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Thomas Kuhn examined the history of science to identify patterns of activity that shape the progress of science.

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- Similar ideas are applicable to social sciences as well, where a social reality can be viewed by different people in different ways, which may constrain their thinking and reasoning about the observed phenomena.
 - For instance, think of politics: People will have a different perception of the role of governments in people's lives, depending on whether they are either conservatives or liberals, and, thus, have different opinions on how to solve social issues.

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- **Paradigms** are often hard to catch because they are *implicit, assumed, and taken for granted*.
- However, recognizing them is key to making sense of and reconciling differences in people's perception of the same social phenomena.
 - **Example:**
 - Liberals believe that the best way to improve secondary education is by hiring more teachers, while conservatives believe that privatizing education is more effective in achieving the same goal.
 - **Why?**
 - Liberals believe more in labor, while conservatives place more faith in competitive markets.

Another Example:

- Why was a certain technology successfully implemented in one organization but failed in another?
 - Social scientists might look at this question from:
 - a **rational lens**: e.g., looking for a rational explanation of the problem, such as inadequate technology
 - a **social lens**: e.g., looking at social deficiencies such as inadequate user training
 - a **political lens**: e.g., looking at instances of organizational politics that may subvert the technology implementation process.
 - Hence, subconscious **paradigms** *often constrain* the concepts that researchers attempt to measure, their observations, and their subsequent interpretations of a phenomenon.
 - However, given the complex nature of a social phenomenon, it is possible that *all of the above* **paradigms** are *partly correct*, and that a fuller understanding of the problem may require an understanding and application of multiple **paradigms**.

Two popular paradigms among social scientists are:

- **Positivism (until mid-20th century):**

- Science or knowledge creation should be *restricted* to *what can be observed* or measured.
- Positivism relies *exclusively on theories that can be tested*.
- This led to *empiricism* and *blind faith* in *observed data* and a rejection of any attempt to extend or reason beyond observable data.

- **Post-Positivism (or postmodernism; mid-late 20th century):**

- Argues that *one can make reasonable inferences about a phenomenon* by *combining empirical observations* with *logical reasoning*.
- They view science not as certain but *probabilistic* (based on many contingencies) and often seek to explore these contingencies to understand the social reality better.
 - Camps: Subjectivists vs. Critical Realists

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- 2. Differentiate between ontology and epistemology.

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- **Ontology:** refers to our assumptions about *how we see the world*
 - E.g., does the world consist mostly of social order or constant change?
- **Epistemology:** refers to our assumptions about *the best way to study the world*
 - E.g., should we use an objective or subjective approach to study social reality?

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- Using these two sets of assumptions, we can categorize social science research as belonging to one of four categories:



Figure 3.1. Four paradigms of social science research
(Source: Burrell and Morgan, 1979)

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- 3. Describe the difference between inductive and deductive research.

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- Scientific research in social sciences operates on two levels: a *theoretical* and an *empirical* level.
 - The *theoretical* level is concerned with abstract concepts about a natural or social phenomenon and relationships between those concepts (i.e., build "theories").
 - The *empirical* level is concerned with testing the theoretical concepts and relationships to see how well they reflect our observations of reality, with the goal of building better theories.
- Both theory and observations are essential components of scientific research.

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Scientific inquiry may take one of two possible forms:

- *Inductive research:*

- The goal is to infer theoretical concepts and patterns from observed data.
 - *Theory-building* research

- *Deductive research:*

- The goal of the researcher is to test concepts and patterns known from theory using new empirical data.
 - *Theory-testing* research (aim is to refine, improve, and extend a theory)

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Theory-building (inductive research) and theory-testing (deductive research) are both critical for the advancement of science.

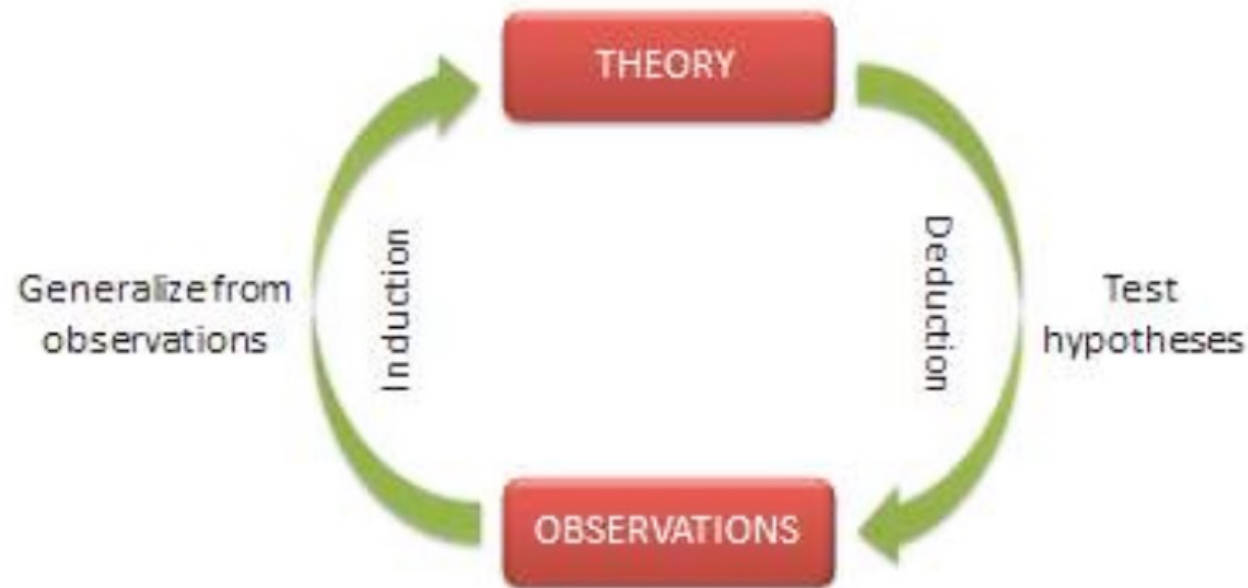


Figure 1.1. The Cycle of Research

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Theory-building (inductive research) and theory-testing (deductive research) are both critical for the advancement of science.

Inductive research is more valuable when there are few prior theories or explanations ...

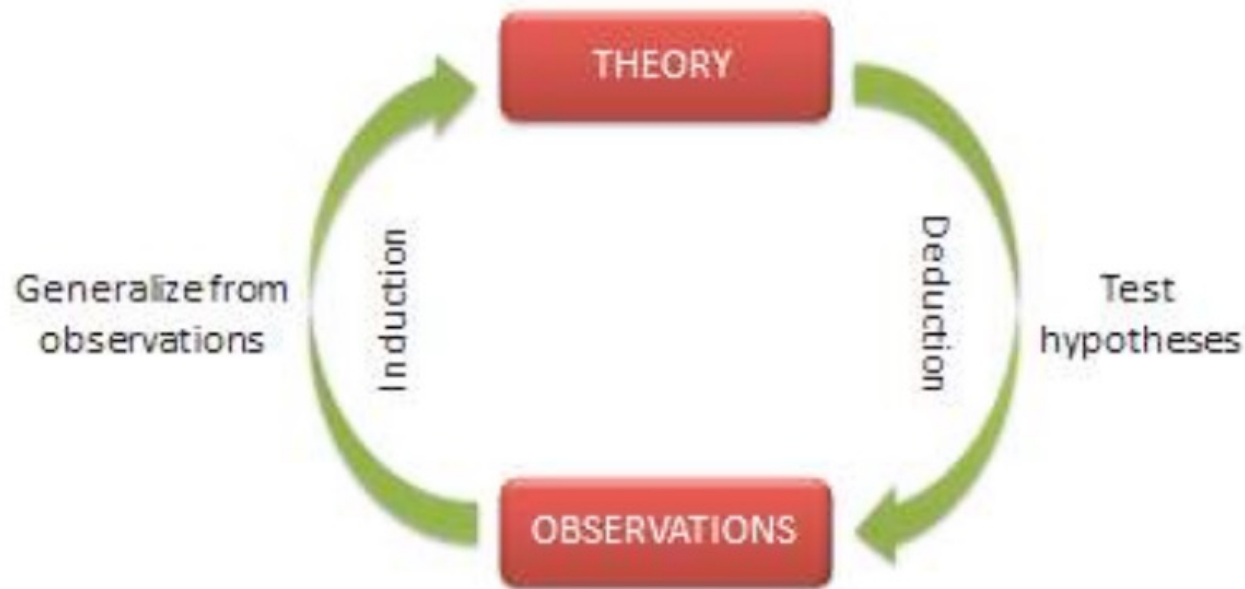


Figure 1.1. The Cycle of Research

Deductive research is more productive when there are many competing theories of the same phenomenon, and researchers are interested in knowing which theory works best and under what circumstances.

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Conducting research requires two skills, needed to operate in the *theoretical* level and *empirical* level respectively:

- *Methodological skills*: “Know-how” (acquired throughout higher education).
- *Theoretical skills*: “Know-what” (requires years of observation and reflection, and are tacit skills that cannot be ‘taught’ but rather learned through experience).

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- 4. Describe the difference between exploratory research, descriptive research and explanatory research using examples.

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- *Exploratory research:*

- is often conducted in new areas of inquiry, where the goals of the research are:
 - 1) To scope out the magnitude or extent of a particular phenomenon, problem, or behavior;
 - 2) To generate some initial ideas ('hunches') about that phenomenon; or
 - 3) To test the feasibility of undertaking a more extensive study regarding the phenomenon.

Example: Citizens' dissatisfaction with governmental policies during an economic recession

- Examination of publicly reported figures (GDP, unemployment, and consumer price index); interviews of experts, economists, or government officials; study of historical examples of dealing with similar issues.

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- **Descriptive research:**

- is directed at making careful observations and detailed documentation of a phenomenon of interest.
- These observations must be based on the scientific method (i.e., logical, confirmable, repeatable, and scrutinizable), and therefore, are more reliable than casual observations by untrained people.

- **Example:**

Chronicling the persistence or evolution of cultural, religious, or ethnic practices in selected communities

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- Explanatory research:

- Seeks explanations of observed phenomena, problems, or behaviors.
 - While *descriptive research* examines the *what, where, and when* of a phenomenon, *explanatory research* seeks answers to *why* and *how* types of questions.
 - It attempts to "connect the dots" in research by identifying causal factors and outcomes of the target phenomenon.
- Example:
 - Understanding the reasons behind adolescent crime or gang violence, with the goal of prescribing strategies to overcome such societal ailments.
 - This requires strong theoretical and interpretation skills, along with intuition, insights, and personal experience.

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- 5. Describe the term unit of analysis, providing examples.

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- The *unit of analysis* refers to the *person, collective, or object* that is the target of the investigation.

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- The *unit of analysis* refers to the *person*, *collective*, or *object* that is the target of the investigation.
 - For instance, if we are interested in studying people's shopping behavior, their learning outcomes, or their attitudes to new technology, then the *unit of analysis* is the *individual*.

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- The *unit of analysis* refers to the *person*, *collective*, or *object* that is the target of the investigation.
 - If we want to study characteristics of street gangs or teamwork in organizations, then the *unit of analysis* is the *group*.

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- The *unit of analysis* refers to the *person*, *collective*, or *object* that is the target of the investigation.
 - For instance, if researchers are interested in understanding how to make web pages more attractive to its users, then the *unit of analysis* is a *web page* (and not users).

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- The *unit of analysis* refers to the *person, collective, or object* that is the target of the investigation.
- Understanding the *unit of analysis* is important because it shapes what type of data you should collect for your study and who you collect it from.
 - E.g., if your unit of analysis is a web page, you should be collecting data about web pages from actual web pages, and not surveying people about how they use web pages.
 - E.g., if your unit of analysis is the organization, then you should be measuring organizational-level variables such as organizational size, revenues, hierarchy, or absorptive capacity.

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- 6. Describe the terms concept, unidimensional construct, multi-dimensional construct.

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- Most research tends to be of the *explanatory type*.
- Explanations require the development of **concepts** or generalizable properties or characteristics associated with objects, events, or people.
 - While objects such as a person, a firm, or a car are not concepts, their specific characteristics or behavior, such as a person's attitude toward immigrants, a firm's capacity for innovation, and a car's weight, can be viewed as concepts.

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- A **construct** is an **abstract concept** that is specifically chosen (or 'created') to explain a given phenomenon.
 - It may be simple, such as a person's *weight*, or a combination of a set of related concepts, such as a person's *communication skill*, which may consist of several underlying concepts, such as the person's *vocabulary*, *syntax*, and *spelling*.

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 - The former instance (weight) is a *unidimensional construct*, while the latter (communication skill) is a *multi-dimensional construct* (i.e., it consists of multiple underlying concepts).

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- 7. How are operational definitions different from dictionary definitions.

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- Constructs used for scientific research **must have** precise and clear definitions that others can use to understand exactly *what it means* and *what it does not mean*.

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- There are two types of definitions:
- **Dictionary definitions:** a construct is often defined in terms of a synonym
 - e.g., attitude may be defined as a disposition, a feeling, or an affect, and an affect in turn is defined as an attitude. (circular; not very useful for elaborating the meaning and content of that construct)
- **Operational definitions:** define the constructs in terms of how they will be empirically measured.
 - e.g., the operational definition of a construct such as *temperature* must specify whether we plan to measure temperature in Celsius, Fahrenheit, or Kelvin scale.

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- 8. What is a variable in the context of scientific research. What types of variables have you read about? Give examples

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- A term frequently associated with, and sometimes used interchangeably with, a construct is a *variable*.
- In scientific research, a *variable* is a measurable representation of an abstract construct.
 - Abstract constructs are not directly measurable, and hence, we look for *proxy measures* called *variables*.
- Example: IQ (intelligence quotient) score
 - *Intelligence* is a construct, and *IQ score* is a variable that measures the intelligence construct.

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Constructs are conceptualized at the theoretical (abstract) plane ...

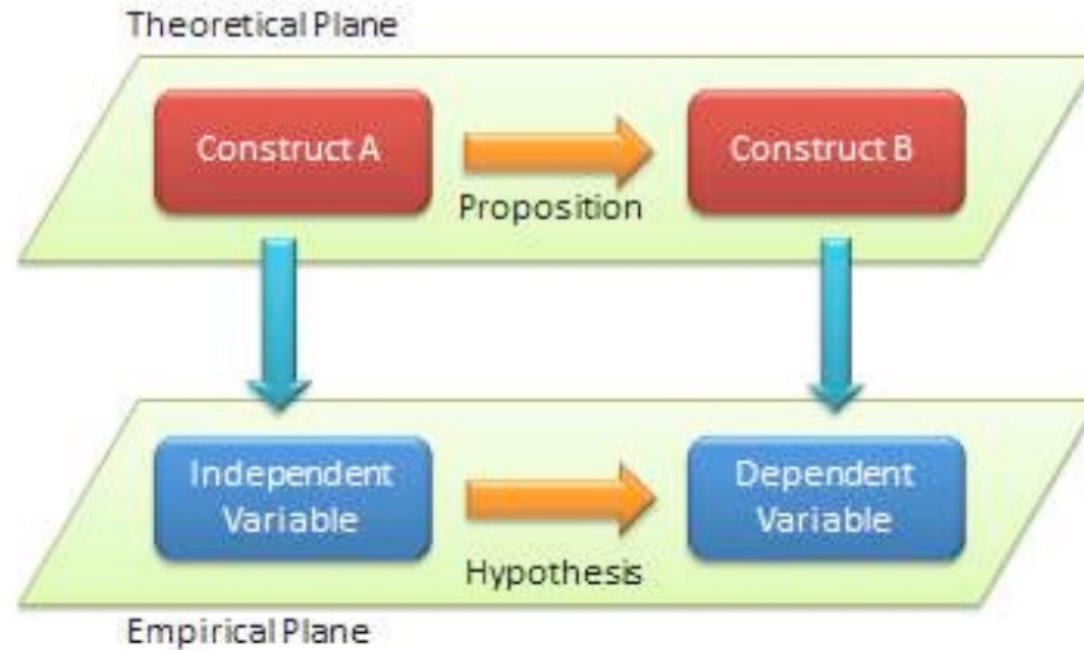


Figure 2.1. The theoretical and empirical planes of research

... while variables are operationalized and measured at the empirical (observational) plane.

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- Depending on their intended use, variables may be classified as independent, dependent, mediating, moderating, or control variables.
 - **Independent variables**: are variables that explain other variables
 - **Dependent variables**: are variables that are explained by other variables
 - **Mediating variables**: are variables that are explained by independent variables while also explaining dependent variables
 - **Moderating variables**: are variables that influence the relationship between independent and dependent variables
 - **Control variables**: are variables that are not pertinent to explaining a given dependent variable, but may have some impact on the dependent variable. These variables must be controlled for in a scientific study, and are therefore called control variables.

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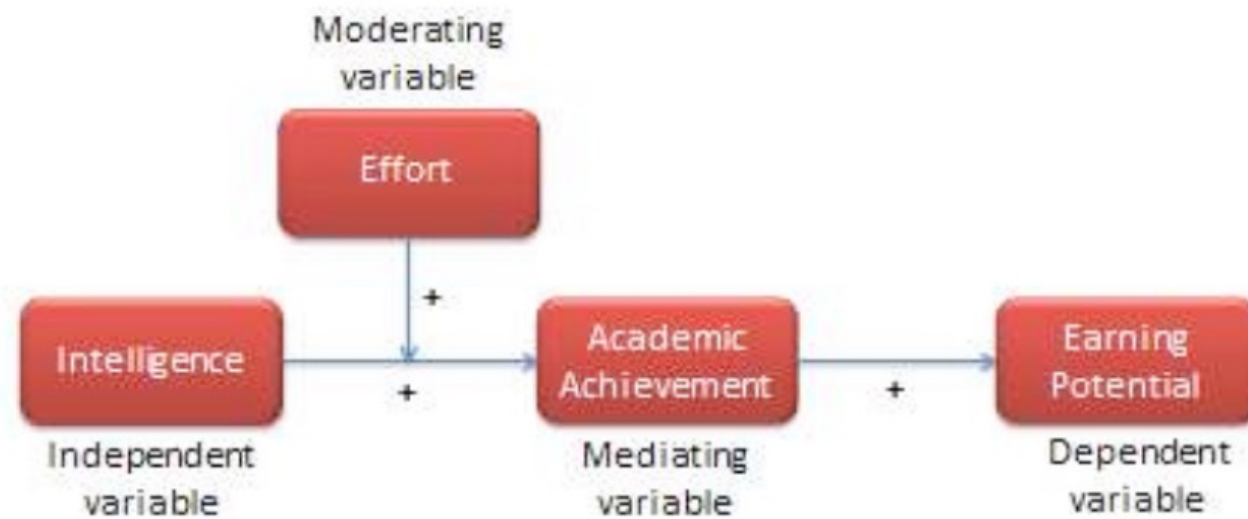


Figure 2.2. A nomological network of constructs

Can you explain the figure above?